

# Dataset shift in neural models

Project: Image-pattern Network Investigation

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Uncertainty and failure analysis

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

---

---

### Similarity and normalisation

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

---

---

### Augmentation and leakage

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

---

---

---

## 2. Practise with fading help

Use W2: Use the right denominator for Dataset shift in neural models. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

---

---

---

## 3. Individual reasoning check

Use the worked case for Dataset shift in neural models. Proposed explanation: “Average the percentages regardless of group sizes”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

---

---

---

---

## 4. Independent transfer case

**Independent case: Analyse a new mixture independently**

A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct. Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

**My independent explanation, trace and limits**

---

---

---

---

---

---

---

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

**5. W1 Audit the two cohorts**

From the table, calculate clear and dim accuracy in each cohort and the clear/dim proportions. Which observed quantities change, and which remain the same?

**My reasoning and evidence**

---

---

---

---

---

---

---

**6. W2 Use the right denominator**

Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop. Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

### My reasoning and evidence

---

---

---

---

---

---

## 7. W3 Separate an alert from an outcome

New-use monitoring shows more dim images but reference labels have not arrived. What can you report now, what can you not calculate, and what checks should precede blaming the neural network?

### My reasoning and evidence

---

---

---

---

---

---

## 8. W4 Analyse a new mixture independently

A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct. Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

### My reasoning and evidence

---

---

---

---

---

---

---

---

## 9. W5 Protect the comparison

A team retrains using new-use errors, then evaluates on those same images and their brightened copies. Explain why the score is not independent evidence and propose a comparison that includes the original model.

### My reasoning and evidence

---

---

---

---

---

---

## 10. Project checkpoint and reflection

Add a labelled dataset shift in neural models evidence sheet to Image-pattern Network Investigation. Show how you can explain how a changed input mix can change overall accuracy. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### My project change, supporting evidence and remaining limit

---

---

---

---

---

---

## 11. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 12. Pictorial case practice: Dataset shift in neural models

Separate within-setting performance from changing setting proportions.

Toy old-use mix: bright 40 cases32 correct; dim 10 cases5 correct. New-use mix: bright 10 cases8 correct; dim 40 cases20 correct.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

## Fixed slice rates, changed pooled accuracy

accuracy percent



Bar length shows the supplied value; the marked baseline is zero.

## Dataset shift in neural models

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Keep slice rates

Bright is 80% in both mixes; dim is 50% in both.



#### 2. Old pooled rate

Old accuracy  $37/50=74\%$ .



#### 3. New pooled rate

New accuracy  $28/50=56\%$ ; more use now occurs in the harder dim setting.

### CHANGE ONE THING

New-use mix returns to 40 bright and 10 dim with the same slice rates.

Explain the changed result using the matching lesson.

Did the supplied model's dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

### My picture-based explanation and changed-case prediction

---

---

---

---

---

---

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

### Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

---

---

---

---

